

Public Health Data Visualization Report

1. Use Case (Problem and Stakeholders)

Public health professionals often work with outbreak datasets that contain a large amount of information. Although these datasets are valuable, they are not always easy to understand without programming skills. Exploring the data usually requires writing R code to filter records and generate new visualizations. This process can be slow and makes it difficult for non-technical users to identify important trends.

For this project, I used the **fluH7N9_china_2013** dataset from the **outbreaks** package in R. This dataset contains information from the first reported human outbreak of avian influenza A(H7N9) in China in 2013. It includes patient age, sex, province, date of symptom onset, and clinical outcome (Recover or Death).

The dataset required preparation before it could be visualized. Unknown ages were recorded as "?", some records contained missing values, and many provinces had only a few reported cases. These issues made the data difficult to interpret directly.

The dashboard was developed for several stakeholders:

- **Public health students** who want to explore outbreak data through interactive visualizations rather than code.
- **Instructors** who need a simple teaching tool for epidemiology and data visualization.
- **Epidemiologists and data analysts** who need a quick overview of the data before performing more advanced statistical analyses.
- **Health officials and non-technical users** who benefit from visual summaries instead of raw datasets.

The main objective was to create an application that makes the outbreak data easier to understand and allows users to explore different patient groups without writing R code.

2. Solution (Pathway to the Solution)

To address this problem, I designed an interactive dashboard using **R Shiny**.

I first explored the dataset and selected five variables that were most useful for understanding the outbreak:

- Age
- Sex
- Clinical outcome
- Province
- Date of symptom onset

These variables allow users to answer common epidemiological questions, such as:

- Which age groups were most affected?
- How were cases distributed across provinces?
- How did the outbreak change over time?
- What is the age distribution of infected patients?

Before creating the dashboard, I cleaned the dataset. Unknown ages (" ? ") were converted to missing values and removed using `drop_na()`. Gender values were recoded into **Male** and **Female**, and clinical outcomes were labelled as **Recover** and **Death**. Patient ages were grouped into four categories (<18, 18–39, 40–59 and 60+) to simplify interpretation. Since many provinces contained only a few cases, they were combined into an **Other** category, while the three most affected provinces (Shanghai, Jiangsu and Zhejiang) remained separate.

The dashboard includes four interactive filters:

- Sex
- Clinical outcome
- Province
- Age range

Whenever users change one of these filters, all visualizations update automatically.

To present the outbreak clearly, I included four different charts:

- **Bar chart:** Number of patients in each age group.
- **Heatmap:** Comparison of age groups across provinces.
- **Epidemic curve:** Cumulative number of cases over time based on symptom onset.
- **Histogram:** Distribution of patient ages.

One issue during development was that some combinations of filters produced no matching records. Instead of displaying empty graphs, I added a validation message that informs users when no data is available for the selected filters. This makes the application more user friendly.

The final dashboard allows users to explore the outbreak quickly and interactively without modifying the R code.

3. Implementation

The project was implemented in **R** using **R Shiny**.

Packages Used

- **shiny** – creates the interactive web application.
- **ggplot2** – generates all visualizations.
- **dplyr** – filters and transforms the dataset.
- **tidyr** – removes missing values.
- **outbreaks** – provides the H7N9 dataset.

Data Preparation

The application loads the **fluH7N9_china_2013** dataset directly from the **outbreaks** package.

Several preprocessing steps were performed:

- Converted unknown ages (" ? ") to **NA**.
- Removed records with missing age or gender.
- Converted gender values into **Male** and **Female**.
- Converted outcome values into **Recover** and **Death**.
- Created four age groups using the **cut()** function.
- Grouped all provinces except Shanghai, Jiangsu and Zhejiang into **Other**.

User Interface

The dashboard uses a **sidebarLayout()**.

The sidebar contains:

- **selectInput()** for Sex
- **selectInput()** for Clinical Outcome
- **selectInput()** for Province
- **sliderInput()** for Age Range

The main panel uses **tabsetPanel()** to organize four visualizations:

- Cases by Age Group

- Province Heatmap
- Epidemic Curve
- Age Distribution

Server Logic

A single `reactive()` dataset filters the data according to the selected user inputs.

Each graph is created using a separate `renderPlot()` function that listens to the reactive dataset. Whenever a filter changes, the plots update automatically without reloading the application.

The application also checks whether the filtered dataset is empty. If no records are available, a message is displayed instead of an empty graph.

Published Application

Shiny App:

<https://github.com/Navidhassanabadi/Digital-Health---Influenza-H7N9-in-China-2013.git>

Submitted Files

The following files are included with this submission:

- `app.R`
- Dataset (fluH7N9_china_2013 from the **outbreaks** package)